

Define CS, fit into CC

1. CUSTOMER SEGMENT(S)CS

The main customer segments of the Automobile Insurance Fraud Detection System include insurance companies, insurance agents, investigators, policyholders, and system administrators. Insurance companies use the system to detect fraudulent claims and reduce financial losses. Insurance agents and investigators analyze claim data to identify suspicious activities, while policyholders benefit from faster claim processing and transparent claim status updates. System administrators manage and monitor the overall system operations.

6. CUSTOMER CONSTRAINTS

- Internet connection required
- Limited technical knowledge of some users
- Data privacy and security concerns
- Need for accurate claim information
- System availability and server uptime
- Device compatibility (mobile/web)
- Processing time for fraud analysis

5. AVAILABLE SOLUTIONSAS

- Manual Claim Investigation – Insurance agents manually verify claim details and documents.
- Rule-Based Fraud Detection Systems** – Systems that detect fraud using predefined rules and conditions.
- Data Analytics Tools** – Analyze historical claim data to find suspicious patterns.
- Machine Learning Models** – Use algorithms to predict whether a claim is fraudulent or genuine.

Explore AS, differentiate

Focus on J&P, tap into BE, understand RC

2. JOBS-TO-BE-DONE / PROBLEMSJ&P

The Automobile Insurance Fraud Detection System aims to assist insurance companies in identifying fraudulent claims using data analysis and machine learning techniques. The system helps insurance agents and investigators analyze claim information efficiently while allowing customers to submit claims easily and track their status. The current challenges include increasing insurance fraud, manual claim verification, large volumes of claim data, delayed claim processing, and high investigation costs.

9. PROBLEM ROOT CAUSERC

- Manual claim verification leads to human errors and slow processing.
- Lack of advanced technology** to detect fraud patterns automatically.
- Large volume of insurance claims** makes fraud difficult to identify.
- Incomplete or incorrect claim information** from users.
- Limited data analysis tools** in traditional systems.
- Weak monitoring of claim history and patterns.**

7. BEHAVIOURBE

- User enters claim details in the system.
- System validates the input data** (vehicle, accident, claim amount).
- Fraud detection model analyzes the claim data.**
- System calculates fraud risk score** based on patterns.
- Suspicious claims are flagged** for further investigation.
- Insurance agent reviews the result** and makes the final decision.

Focus on J&P, tap into BE, understand RC

Identify strong TR & EM

3. TRIGGERSTR

- User Registration Trigger** – Activated when a new user creates an account in the system.
- User Login Trigger** – Starts when a user logs into the application.
- Claim Submission Trigger** – Occurs when a customer submits a new insurance claim.
- Fraud Detection Trigger** – Activated when the system analyzes claim data using the fraud detection model.
- Suspicious Claim Trigger** – Triggered when the system detects a high fraud risk score.
- Claim Review Trigger** – Activated when an insurance agent reviews a claim.

10. YOUR SOLUTIONSL

- Machine Learning–Based Fraud Detection: Use algorithms to analyze claim data and identify suspicious patterns.
- Automated Claim Analysis:** The system automatically evaluates claims and generates a fraud risk score.
- Real-Time Detection:** Fraudulent claims can be detected quickly during claim submission.
- Data Validation:** Verify claim details such as vehicle information, accident details, and claim amount.
- Fraud Alerts:** Notify insurance agents when a claim is highly suspicious.
- Dashboard & Reports:** Provide visual reports and statistics for

8. CHANNELS of BEHAVIOURCH

- Channels – Automobile Insurance Fraud Detection System (Short)
- Web Application – Users submit and track insurance claims online.
 - Mobile Application – Customers and agents access the system through smartphones.
 - Email Notifications – System sends alerts for claim status and fraud detection updates.
 - Admin Dashboard – Administrators monitor claims and fraud reports.
- Behaviour – Automobile Insurance Fraud Detection System (Short)
- User submits claim details through web or mobile app.
 - System validates the input data (vehicle and accident information).
 - Fraud detection model analyzes the claim and calculates risk score.

Identify strong TR & EM

4. EMOTIONS: BEFORE / AFTER



Emotions Before Using the System

- Frustration due to slow claim verification
- Stress about fraudulent claims
- Confusion while handling large amounts of claim data
- Lack of trust in manual claim investigation
- Delay in claim approval process

Emotions After Using the System

- Confidence in fraud detection results
- Satisfaction due to faster claim processing
- Reduced workload for insurance agents
- Better trust in automated analysis

- Suspicious claims are flagged automatically.
- Insurance agent reviews the claim and takes the final decision.